

CT, MRI, PET, SPECT, Ultrasound, Optical imaging

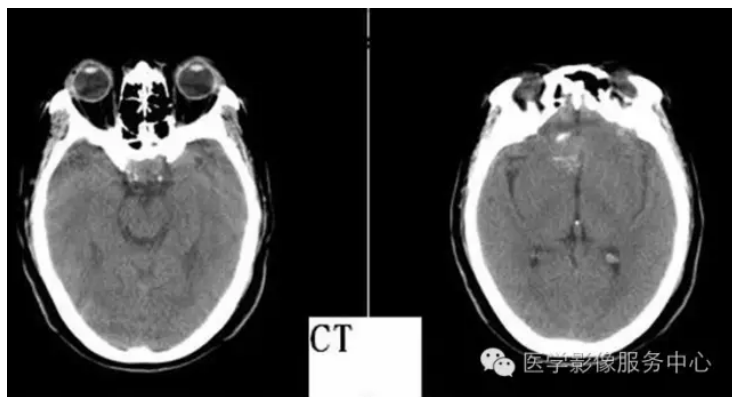
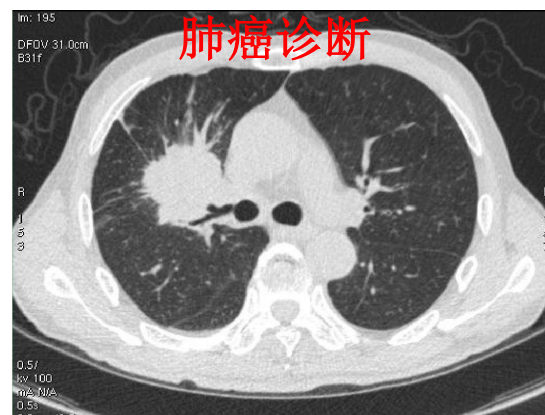
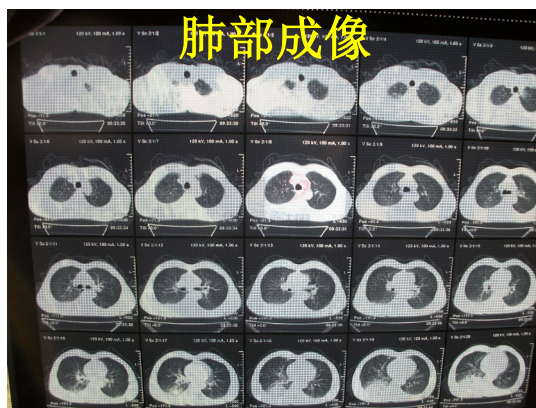
Table 1. Comparison of Some Key Imaging Modalities (Adapted with Permission from Refs 33 and 34; Copyright 201

modality	spatial resolution	depth limit	time	target ^a	imaging agents	probe sensitivity (M)
CT	50–200 μm	none	seconds–minutes	A, P	heavy elements, e.g., iodine and barium	not well-characterized
MRI	25–100 μm	none	minutes–hours	A, P, M	superparamagnetic: Fe_3O_4 , Fe_2O_3 ; paramagnetic: Gd, Mn	10^{-3} – 10^{-5}
PET and SPECT	1–2 mm	none	minutes	P, M	radionuclides PET: ^{18}F , ^{13}N , ^{11}C , ^{15}O , ^{64}Cu , ^{68}Ga , ^{89}Zr , ^{124}I ; SPECT: $^{99\text{m}}\text{Tc}$, ^{111}In	PET: 10^{-11} – 10^{-12} ; SPECT: 10^{-10} – 10^{-11}
ultrasound	50–500 μm	none	seconds–minutes	A, P	microbubbles	not well-characterized
optical imaging	1–5 mm	mm	seconds–minutes	P, M	fluorescent dyes or quantum dots	bioluminescence: 10^{-15} – 10^{-17} ; fluorescence 10^{-9} – 10^{-12}

^aA, anatomical; P, physiological; M, molecular.

	probe sensitivity (M)	advantages	disadvantages
CT	not well-characterized	high-resolution; relatively lower cost; no depth limit	poor sensitivity; not functional (contrast agents are required for tissue differentiation); radiation
MRI	10^{-3} – 10^{-5}	high-resolution; no radiation; functional imaging with modification (fMRI); no depth limit	high cost; low sensitivity; not functional; limited availability for patients with metallic implants
PET, SPECT	PET: 10^{-11} – 10^{-12} ; SPECT: 10^{-10} – 10^{-11}	high sensitivity; functional imaging; no depth limit	low resolution; high cost; radiation
Ultrasound	not well-characterized	low cost; ease of operation; no radiation	low resolution; limited imaging depth
Optical imaging	bioluminescence: 10^{-15} – 10^{-17} ; fluorescence 10^{-9} – 10^{-12}	high sensitivity; no radiation; multichannel imaging	not for clinical imaging; low resolution; limited imaging depth

CT (Computed Tomography)，即**电子计算机断层扫描**，它是利用精确准直的X线束、 γ 射线、超声波等，与灵敏度极高的探测器一同围绕人体的某一部位作一个接一个的断面扫描，具有扫描时间快，图像清晰等特点，可用于多种疾病的检查；根据所采用的射线不同可分为：X射线CT（X-CT）、超声CT（UCT）以及 γ 射线CT（ γ -CT）等。

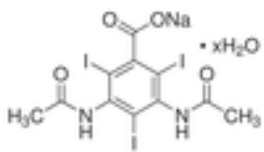


颅脑CT

Contrast agents of CT

Ionic contrast media typically, but not always, have higher osmolality and more side-effects
Commonly used iodinated contrast agents

Compound	Name	Type	Iodine content	Osmolality
Ionic	diatrizoate (Hypaque 50)	Monomer	300 mgI/ml	1550 High
Ionic	metrizoate (Isopaque 370)	Monomer	370 mgI/ml	2100 High
Ionic	ioxaglate (Hexabrix)	Dimer	320 mgI/ml	580 Low



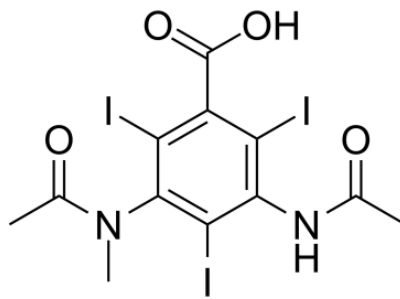
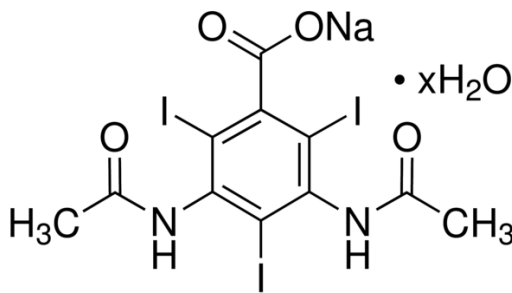
泛影酸钠,水合物

Sodium diatrizoate hydrate

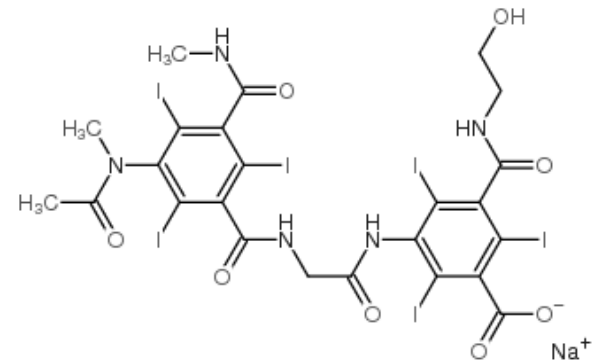
分子式: $C_{11}H_8I_3N_2NaO_4 \cdot xH_2O$

CAS号: 737-31-5

MDL号: MFCD00051544



Metrizonic acid



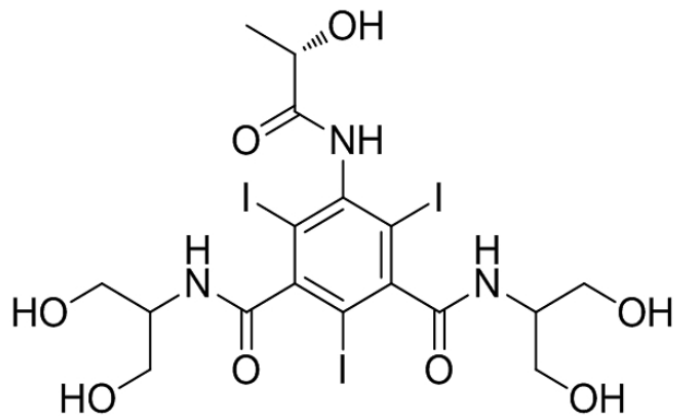
碘克沙酸钠
Ioxaglate Sodium

Contrast agents of CT

Non-ionic contrast media have lower osmolality and tend to have fewer side-effects.

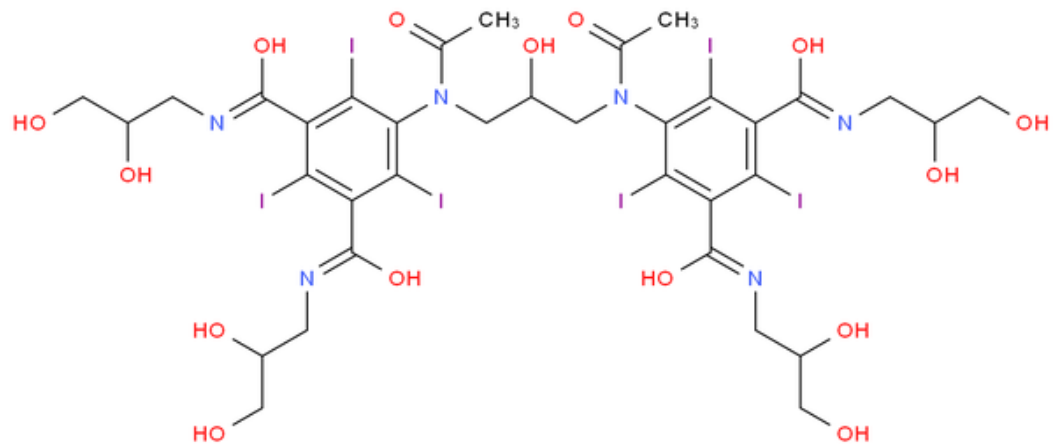
Commonly used non-ionic contrast agents

Compound	Name	Type	<u>Iodine content</u>	Osmolality
Non-ionic iopamidol (Isovue 370)		Monomer	370 mgI/ml	796 Low
Non-ionic iohexol (Omnipaque 350)		Monomer	350 mgI/ml	884 Low
Non-ionic ioxilan (Oxilan 350)		Monomer	350 mgI/ml	695 Low
Non-ionic iopromide (Ultravist 370)		Monomer	370 mgI/ml	774 Low
Non-ionic iodixanol (Visipaque 320)		Dimer	320 mgI/ml	290 Low



碘帕醇

iopamidol



Iodixanol, 碘克沙醇

IUPAC: 5-[acetyl-[3-[N-acetyl-3,5-bis(2,3-dihydroxypropylcarbamoyl)-2,4,6-triiodoanilino]-2-hydroxypropyl]amino]-1-N,3-N-bis(2,3-dihydroxypropyl)-2,4,6-triiodobenzene-1,3-dicarboxamide

CAS No.: 92339-11-2

Contrast agents of Barium sulfate

Barium sulfate is mainly used in the imaging of the digestive system. The substance exists as a water-insoluble white powder that is made into a slurry with water and administered directly into the gastrointestinal tract.

Barium enema (large bowel investigation) and DCBE (double contrast barium enema)
Barium swallow (oesophageal investigation)
Barium meal (stomach investigation) and double contrast barium meal
Barium follow through (stomach and small bowel investigation)

Barium sulfate, an insoluble white powder is typically used for enhancing contrast in the GI tract. Depending on how it is to be administered the compound is mixed with water, thickeners, de-clumping agents, and flavourings to make the contrast agent. As the barium sulfate doesn't dissolve, this type of contrast agent is an opaque white mixture. It is only used in the digestive tract; it is usually swallowed or administered as an enema. After the examination, it leaves the body with the feces

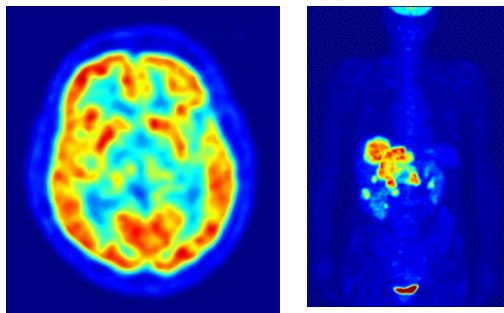
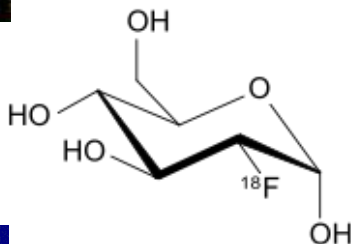
gastrointestinal tract: 胃肠道

Barium enema: 钡剂灌肠; 钡灌肠; 灌肠造影;

Positron Emission Computed Tomography (PET)



正电子发射计算机断层扫描全称为：
正电子发射型计算机断层显像
(Positron Emission Computed Tomography, 简称PET)，是核医学领域比较先进的临床检查影像技术。
正常范围PET特别适用于在没有形态学改变之前，早期诊断疾病，发现亚临床病变以及评价治疗效果。目前，PET在肿瘤、冠心病和脑部疾病这三大类疾病的诊疗中尤其显示出重要的价值。

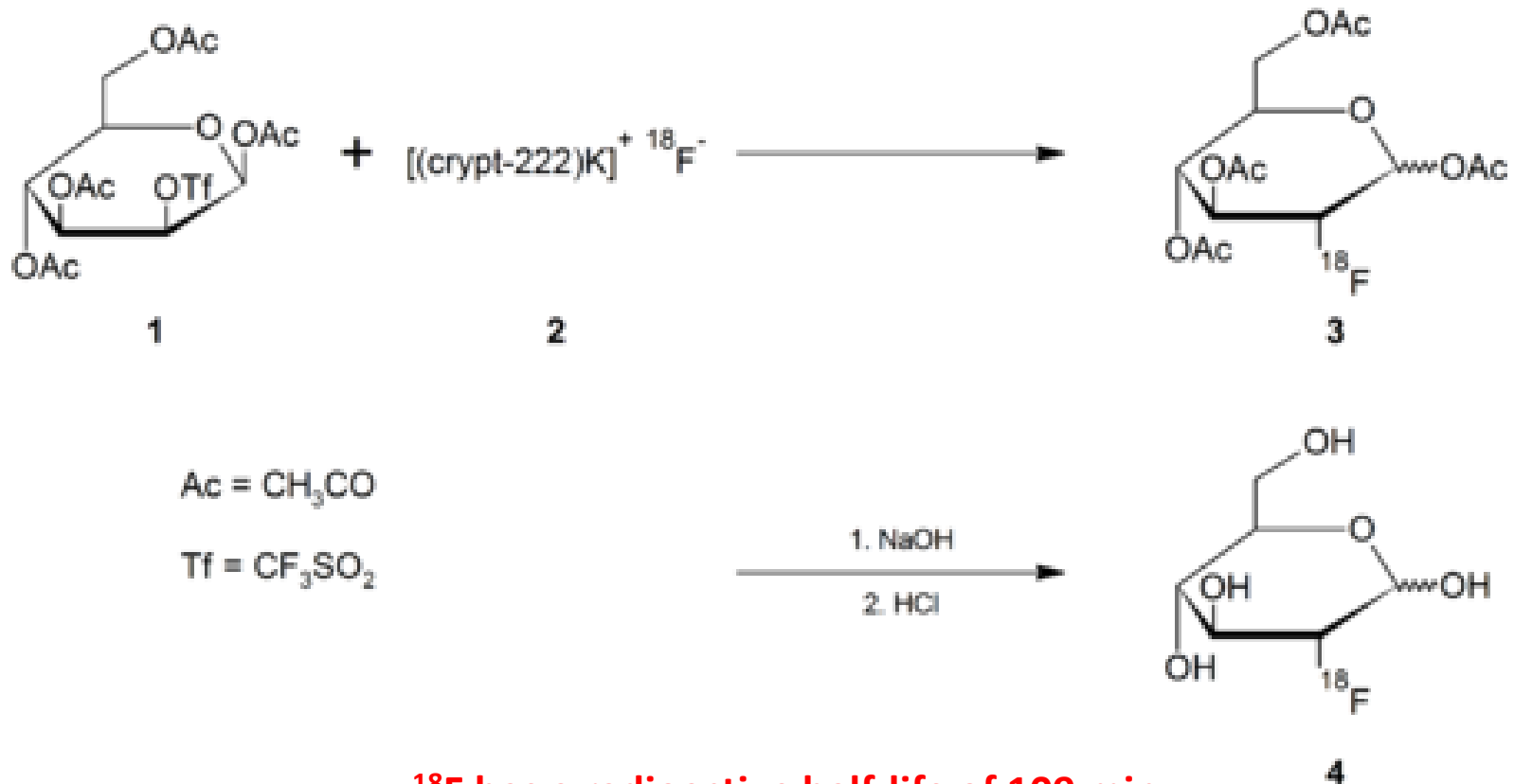


与CT比较, PET 影像的病灶解剖定位和形态观察显得不够精确和细致, 尚不能完全满足外科手术要求。传统解决办法是利用计算机软件将同一病人的CT影像与PET影像进行“图像融合”, 获得含有代谢信息的解剖影像。

Whole-body PET scan using ^{18}F -FDG

PET大致方法是，将某种物质，一般是生物生命代谢中必须的物质，如：葡萄糖、蛋白质、核酸、脂肪酸，**标记上短寿命的放射性核素（如 ^{18}F ， ^{11}C 等）**，注入人体后，通过对于该物质在代谢中的聚集，来反映生命代谢活动的情况，从而达到诊断的目的。

最近各医院主要使用的物质是氟代脱氧葡萄糖，简称**FDG**。其机制是，人体不同组织的代谢状态不同，在高代谢的恶性肿瘤组织中葡萄糖代谢旺盛，聚集较多，这些特点能通过图像反映出来，从而可对病变进行诊断和分析。



^{18}F has a radioactive half-life of 109 min

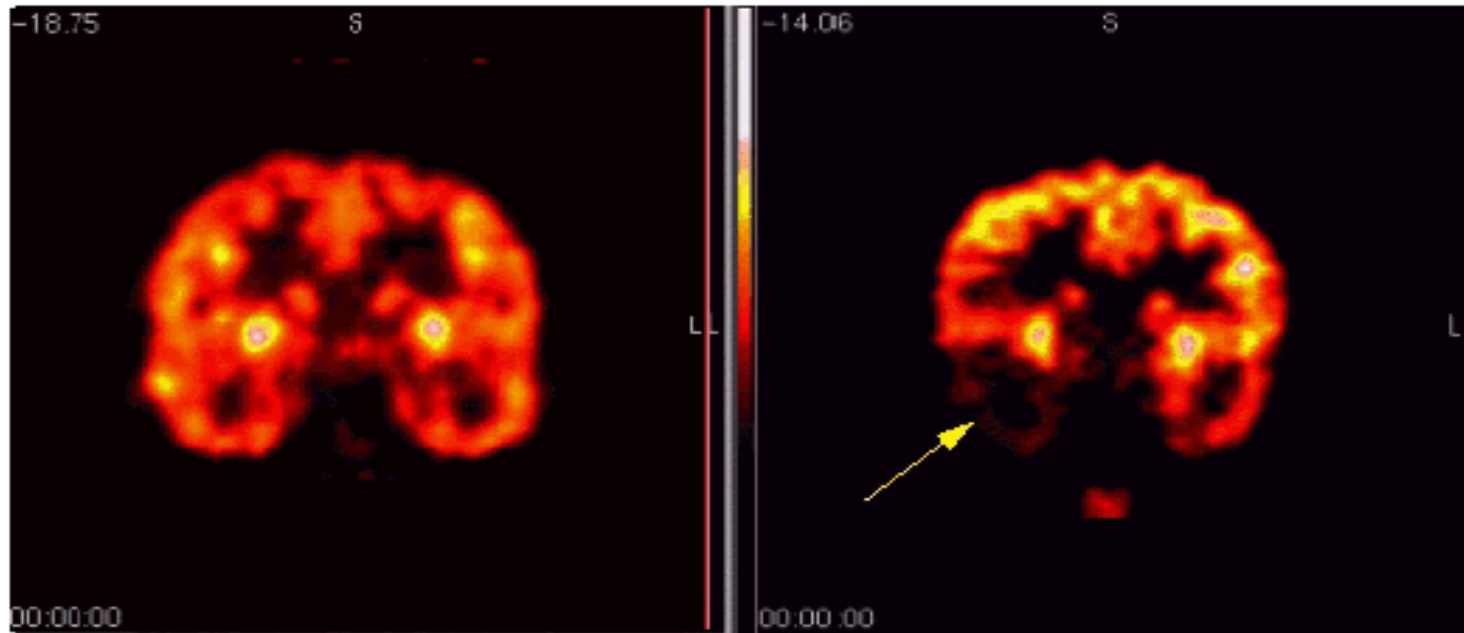
What is PET

- PET is a noninvasive, diagnostic imaging technique for measuring the metabolic activity of cells in the human body.
- It was developed in the mid 1970s and it was the first scanning method to give functional information about the brain.
- -----
- Existence first postulated in 1928 by Paul Dirac
- First observed in 1932 by Carl D. Anderson, who gave the positron its name. He also suggested to rename the electron to “negatron” but he was unsuccessful.

What is a Positron?

- A Positron is an anti-matter electron, it is identical in mass but has an apposite charge of +1.
- Positron can come from different number of sources, but for PET they are produced by nuclear decay.
- Nuclear decay is basically when unstable nuclei are produced in a **cyclotron** by bombarding the target material with protons, and as a result a neutron is released.
 $18\text{-O} + \text{proton} \Rightarrow 18\text{-F} + \text{neutron}$
- In PET the target material is chosen so that the product of the bombardment decays to a more stable state isotope by **emitting a positron**. One of these protons decays into a neutron emitting in the process a positron an a **neutrino**.
$$^{18}\text{F} \rightarrow ^{18}\text{O} + e^{+} + \nu$$
- **Positrons** will continue to interact with surrounding negative electrons and annihilate, producing a pair of photons to γ Energy is emitted in the form of ray, and neutrino hardly interacts with any matter, which can be ignored
- After ^{18}F -FDG decays radioactively, however, its 2'-fluorine is converted to $^{18}\text{O}^{-}$, and after picking up a **proton** H^{+} from a **hydronium ion** in its aqueous environment, the molecule becomes glucose-6-phosphate labeled with harmless nonradioactive "heavy oxygen" in the hydroxyl at the 2' position. The new presence of a 2' hydroxyl now allows it to be metabolized normally in the same way as ordinary glucose, producing non-radioactive end-products.

PET image of brain



Normal brain

PET scans of the brain are used to evaluate patients who have memory disorders of an undetermined cause, suspected or proven brain tumors or seizure disorders that are not responsive to medical therapy and are therefore candidates for surgery.

Image of the brain of a 9 year old female with a history of seizures poorly controlled by medication. PET imaging identifies the area (indicated by the arrow) of the brain responsible for the seizures. Through surgical removal of this area of the brain, the patient is rendered "seizure-free".

Seizures: 癫痫发作

PET image of heart

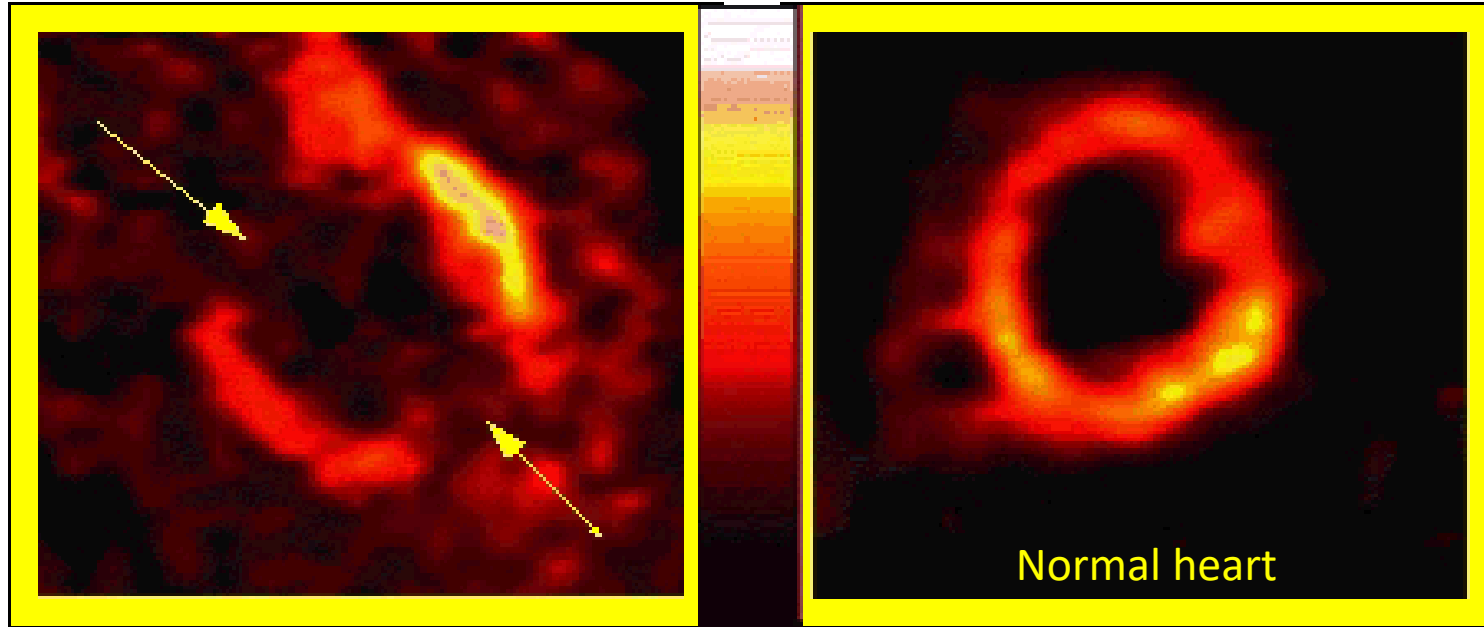


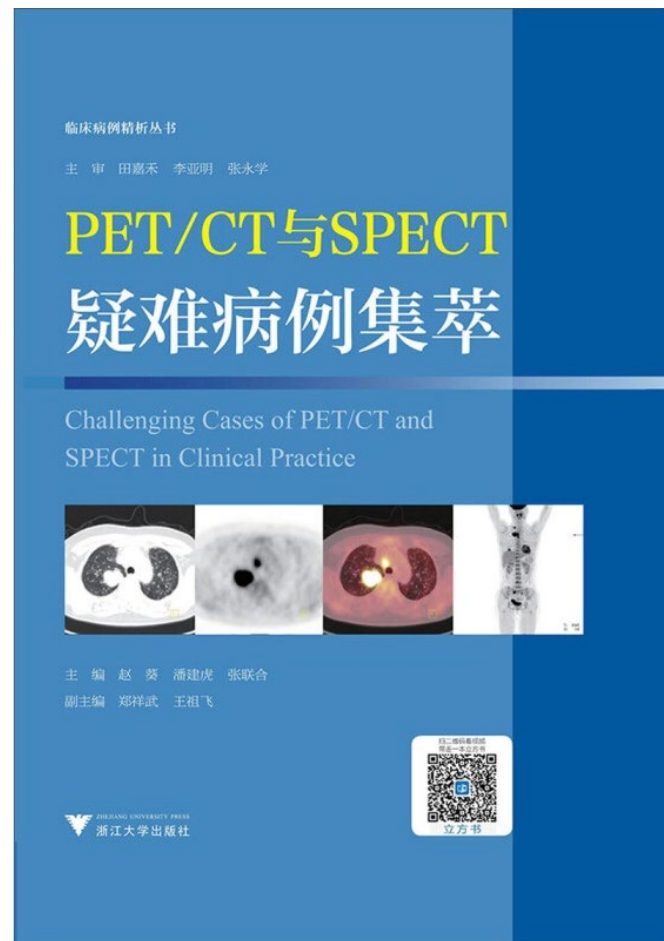
Image of heart which has had a **myocardial infarction (heart attack)**. The arrow points to areas that have been damaged by the attack, indicating "dead" myocardial tissue. Therefore, the patient will not benefit from heart surgery, but may have other forms of treatment prescribed.

Combined with a myocardial perfusion study, PET scans allow differentiation of nonfunctioning heart muscle from heart muscle that would benefit from a procedure, such as coronary bypass for instance.

myocardial infarction: 心肌梗塞

单光子发射计算机断层成像术(Single-Photon Emission Computed Tomography, SPECT)和正电子发射断层成像术(Positron Emission Tomography, PET)是核医学的两种CT技术，由于它们都是对从病人体内发射的 γ 射线成像，故统称发射型计算机断层成像术(Emission Computed Tomography, ECT)。

SPECT的基本本成像原理是：首先病人需要摄入含有**半衰期适当的放射性同位素**药物，在药物到达所需要成像的断层位置后，由于放射性衰变，将从断层处发出 γ 光子，位于外层的 γ 照相机探头的每个灵敏点探测沿一条投影线(Ray)进来的 γ 光子，通过闪烁体将探测到的高能 γ 射线转化为能量较低但数量很大的光信号，通过光电倍增管将光信号转化为电信号并进行放大，得到的测量值代表人体在该投影线上的放射性之和。在同一条直线上的灵敏点可探测人体一个断层上的放射性药物，它们的输出称作该断层的一维投影(Projection)。



Contrast agents of SPECT

FDA-approved radiotracers for brain SPECT scan include

Tc-99m HMPAO (Hexamethyl propyleneamine oxime),

Tc-99m ECD (ethyl cysteinate dimer),

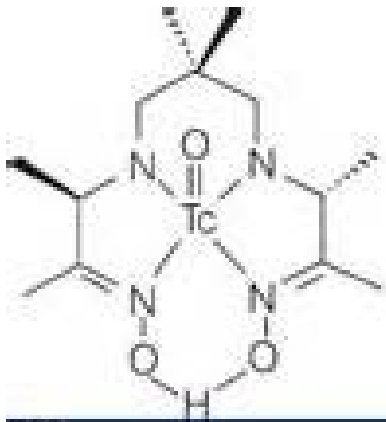
I-123 isopropylidoamphetamine (IMP),

and thallium-201 diethyldithiocarbamate (Tl-DDC).

Tc: 锝，中国大陆称锝 (de'),
港澳称鐽，台湾称鍆。

铊 (Thallium)，化学符号Tl

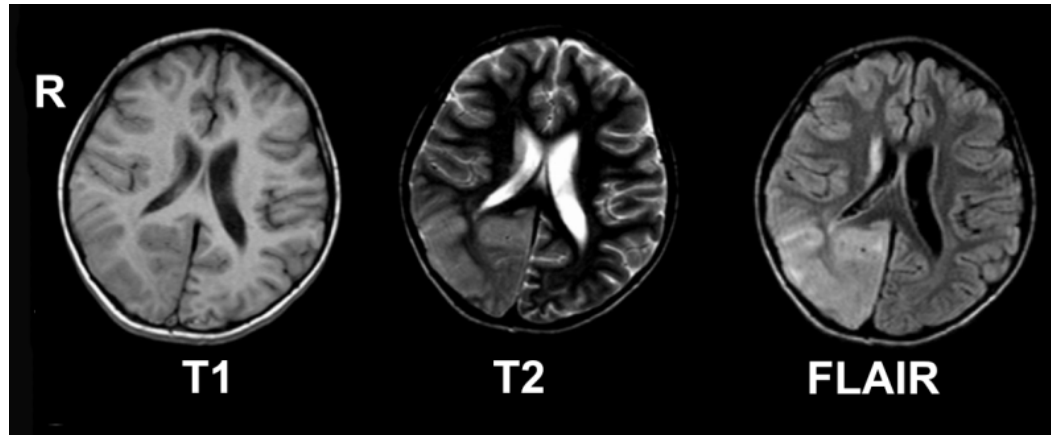
In recent years, combination (hybrid) technologies using SPECT imaging have been developed; **such as SPECT/CT**, which superimposes data from the grey CT and colour-coded SPECT images, and can lead to more accurate diagnoses.



锝 (Technetium)，元素符号Tc，
原子序数为43，原子量为97.9072，
在元素周期表中属VIB族，单质为
银白色金属

Tc99m，半衰期6.02h，纯伽马射线，
单一能量141keV，十分适合于核医学成像

MRI也就是磁共振成像，英文全称是：Magnetic Resonance Imaging。经常为人们所利用的原子核有： ^1H 、 ^{11}B 、 ^{13}C 、 ^{17}O 、 ^{19}F 、 ^{31}P 。在这项技术诞生之初曾被称为核磁共振成像，到了20世纪80年代初，作为医学新技术的NMR成像（NMR Imaging）一词越来越为公众所熟悉。随着大磁体的安装，有人开始担心字母“N”可能会对磁共振成像的发展产生负面影响。另外，“nuclear”一词还容易使医院工作人员对磁共振室产生另一个核医学科的联想。因此，为了突出这一检查技术不产生电离辐射的优点，同时与使用放射性元素的核医学相区别，放射学家和设备制造商均同意把“核磁共振成像术”简称为“磁共振成像（MRI）”。



FLAIR是fluid attenuated inversion recovery的英文缩写，在脑、脊髓MRI（核磁共振）中常用

T1弛豫的发生是因为旋转核与周围环境（即晶格，lattice）之间有能量交换，引起up状态和down状态的原子核数量发生改变，重新恢复到未加B1的平衡状态时的数量分布，因此Mz会恢复到M0，而T1也称为**自旋-晶格弛豫时间**。

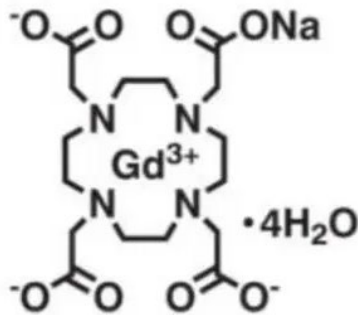
T2弛豫的发生也有一定程度的上述因素，但除此之外，也因为旋转核相互之间有能量交换，各个原子核旋转的相位变得随机，其磁化向量的净值（Mxy）逐渐衰减。故T2也称为**自旋-自旋弛豫时间**。



Contrast agents of MRI

The complex (**Gd-DOTA**) meglumine has recently been used as an MRI contrast agent in humans. Due to its particularly interesting physicochemical properties, the risk of in vivo dissociation of the complex is reduced. Indeed, as a result of the macrocyclic nature of the DOTA ligand, Gd-DOTA appears very stable, demonstrating a calculated conditional stability constant of $10^{22.19}$ at pH = 7.

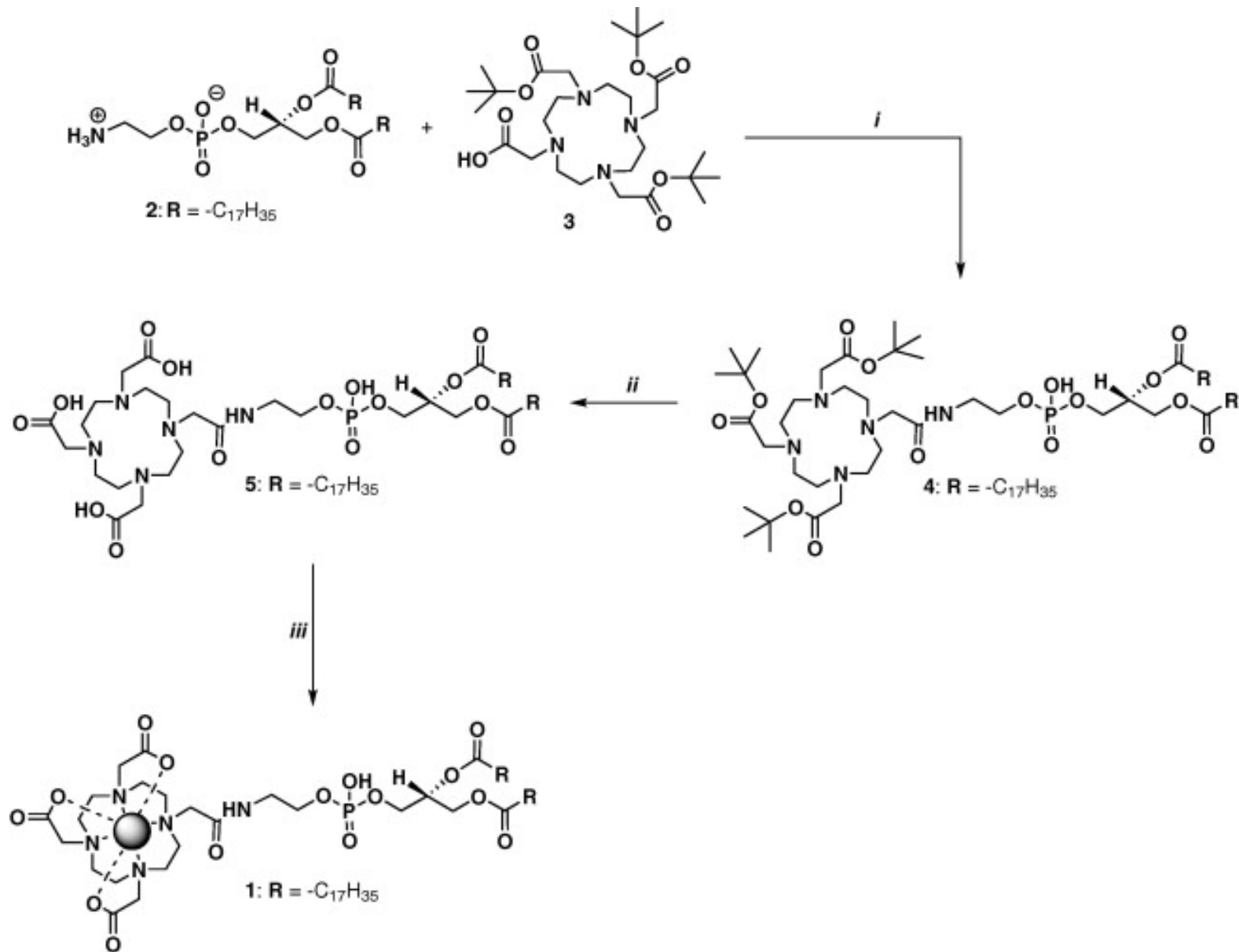
Other characteristics making Gd-DOTA a positively attractive compound include slow dissociation kinetics inherent in the rigidity of the macrocycle and marked specific affinity of DOTA for gadolinium in comparison with other endogenous ions.



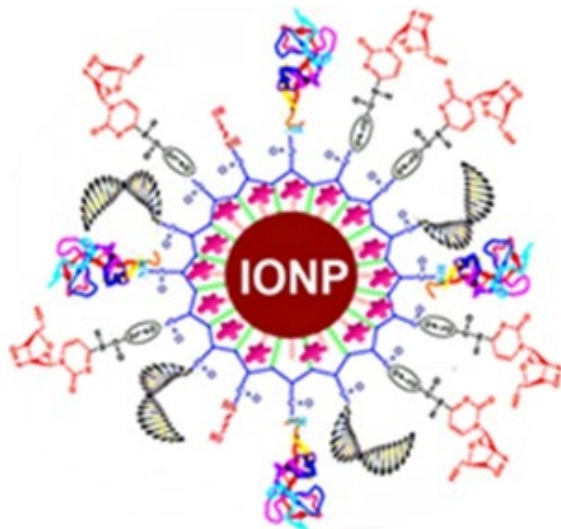
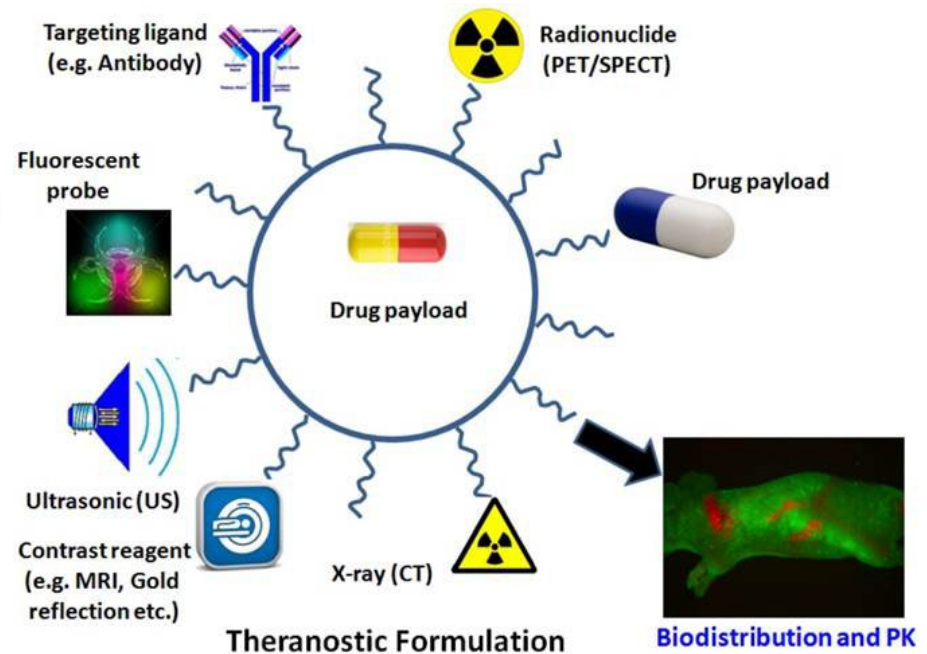
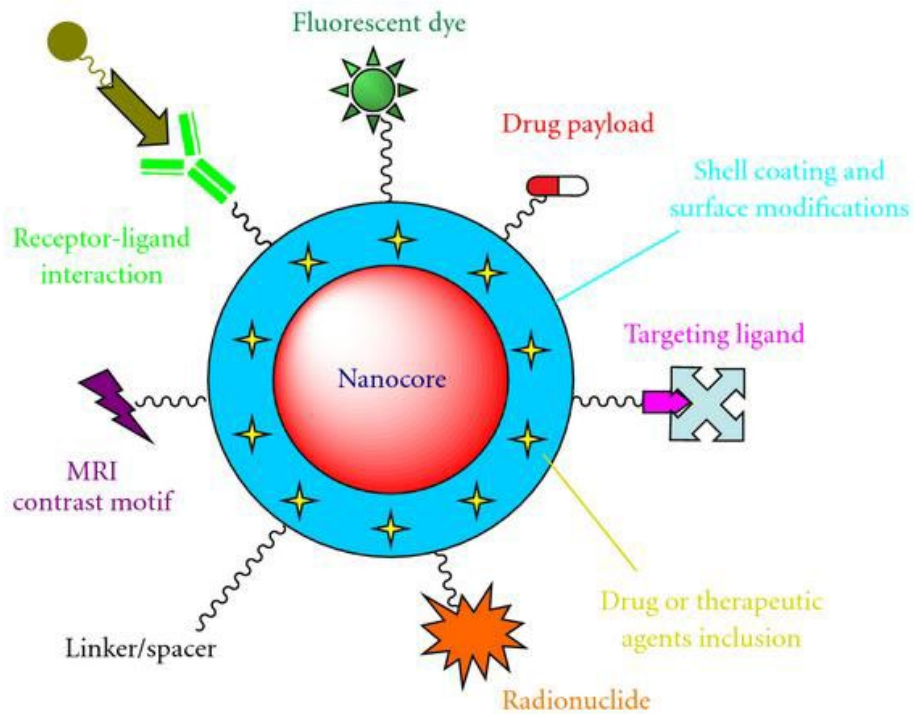
- 1、T1观察解剖结构较好。
- 2、T2显示组织病变较好。
- 3、水为长T1长T2，脂肪为短T1短T2。
- 4、长T1为黑色，短T1为白色。
- 5、长T2为白色，短T2为黑色。
- 6、水T1黑，T2白。
- 7、脂肪T1白，T2灰白。
- 8、T2对出血敏感，因水T2呈白色。

T1加权成像、T2加权成像
所谓的加权就是“突出”的意思
T1加权成像（T1WI）----突出组织T1
弛豫（纵向弛豫）差别
T2加权成像（T2WI）----突出组织T2
弛豫（横向弛豫）差别。

MRI agents



Multifunctional nanoparticles



IONP: Magnetic Iron Oxide Nanoparticle (MRI contrast)

Targeting ligands

- 1). **Hydrophobic drugs (encapsulation)**
- 2). **Hydrophilic drugs (conjugation and protease-activated release)**
- 3). **siRNA-expressing DNA cassette**

Fluorescence imaging:

UV-Vis, NIR

Organic/polymeric:

PD (polymer dot)

QD (quantum dot)

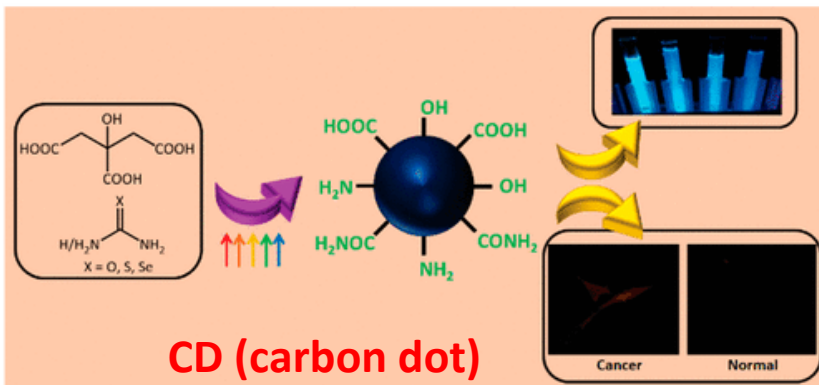
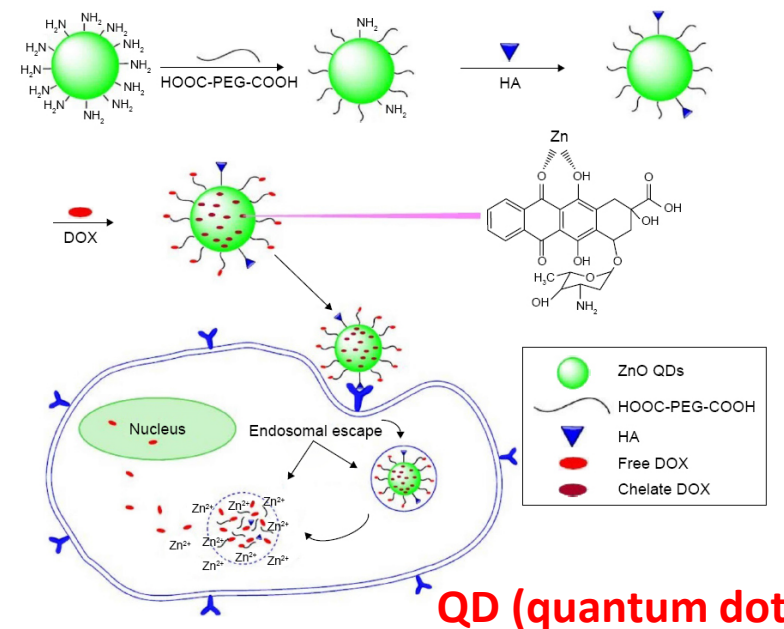
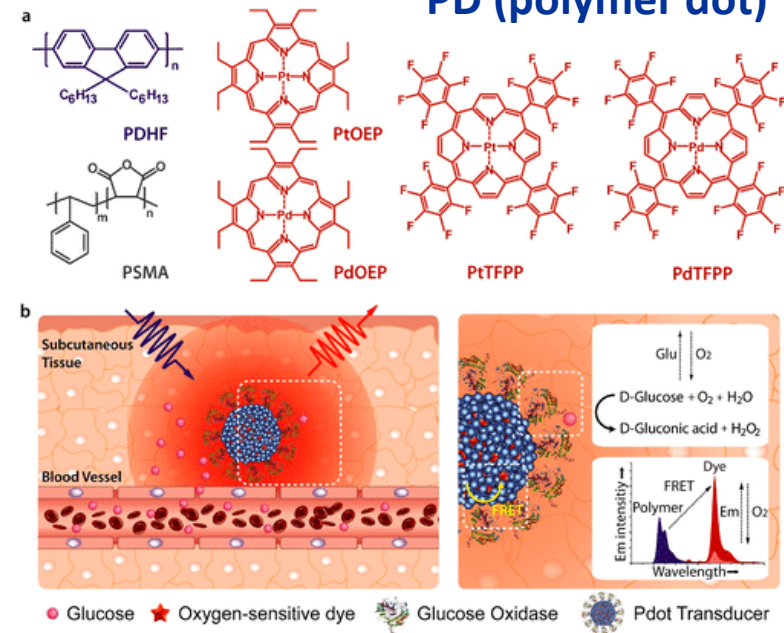
CD (carbon dot)

Upconversion nanoparticles

Two-photon imaging

Photoacoustic imaging

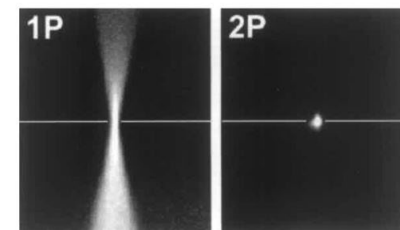
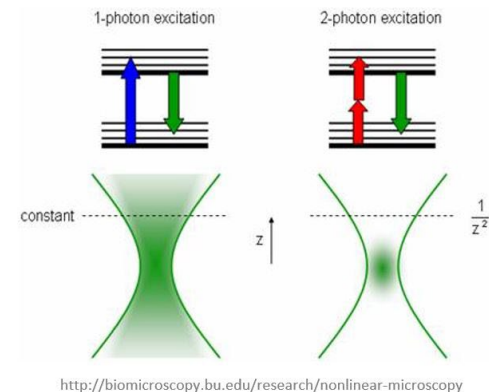
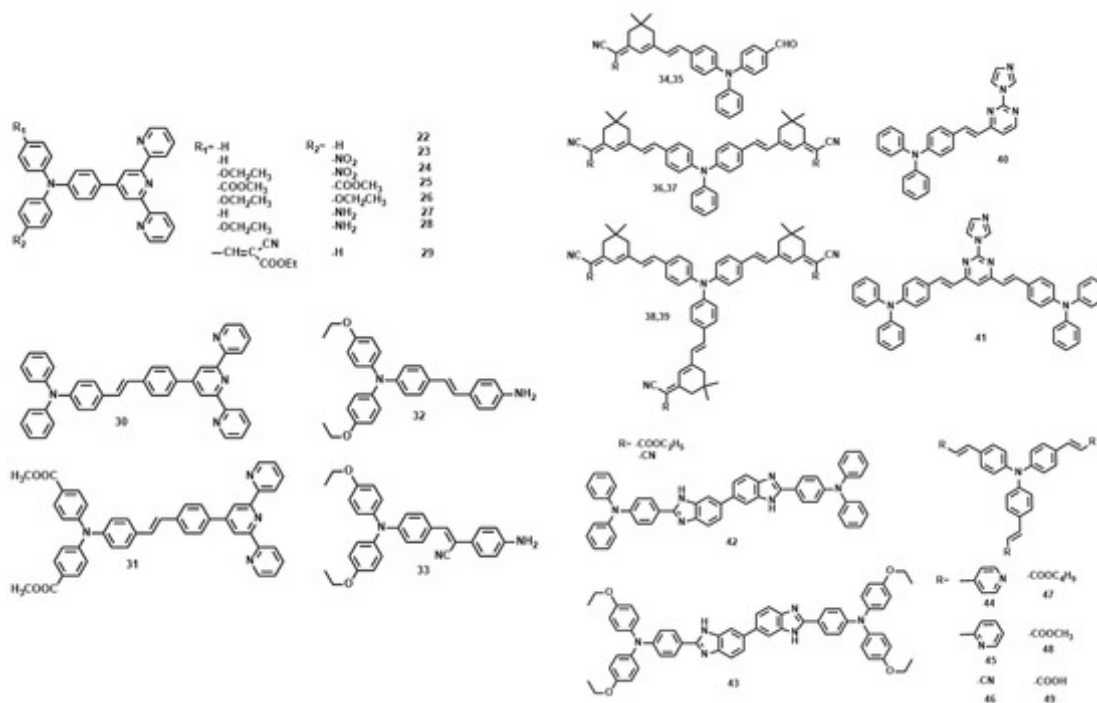
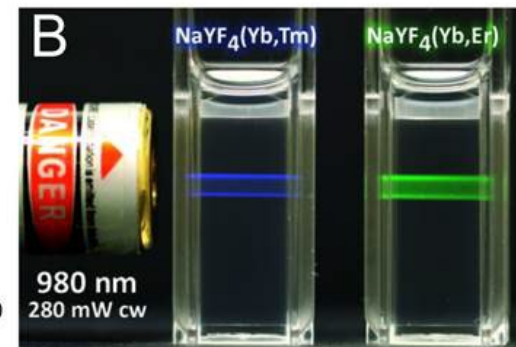
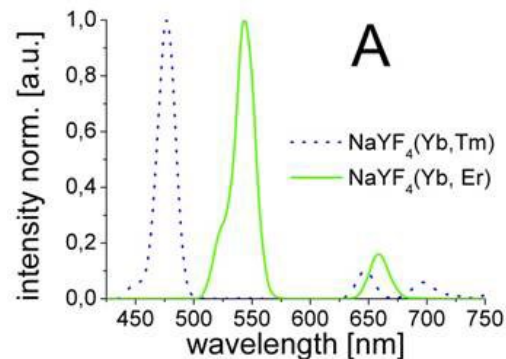
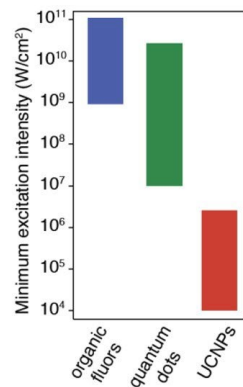
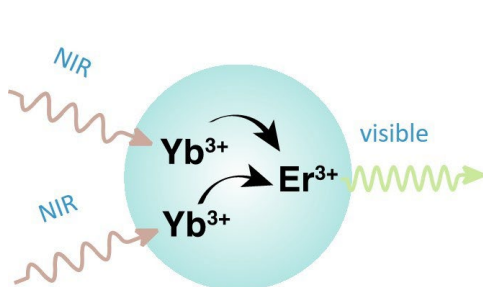
PD (polymer dot)



Graphical representation of the hyaluronic acid-ZnO quantum dots-dicarboxyl-terminated poly(ethylene glycol) (HA-ZnO-PEG) drug delivery system printed with permission from Cai X, Luo Y, Zhang W, et al. pH-Sensitive ZnO quantum dots-doxorubicin nanoparticles for lung cancer targeted drug delivery. *Materials*. 2016;8(34):22442–22450. Copyright 2016 American Chemical Society.¹⁷

Abbreviations: DOX, doxorubicin; HA, hyaluronic acid; PEG, poly(ethylene glycol); QDs, quantum dots.

upconversion and two-photon materials



M. Rubart (2004) Two-Photon Microscopy of Cells and Tissue. Circulation Research 95:1154-1166

光声成像（Photoacoustic Imaging, PAI）是近年来发展起来的一种非侵入式和非电离式的新型生物医学成像方法。当脉冲激光照射到（热声成像则特指用无线电频率的脉冲激光进行照射）生物组织中时，组织的光吸收域将产生超声信号，我们称这种由光激发产生的超声信号为光声信号。生物组织产生的光声信号携带了组织的光吸收特征信息，通过探测光声信号能重建出组织中的光吸收分布图像。光声成像结合了纯光学组织成像中高选择特性和纯超声组织成像中深穿透特性的优点，可得到高分辨率和高对比度的组织图像，从原理上避开了光散射的影响，突破了高分辨率光学成像深度“软极限”（~1 mm），可实现50 mm的深层活体内组织成像。

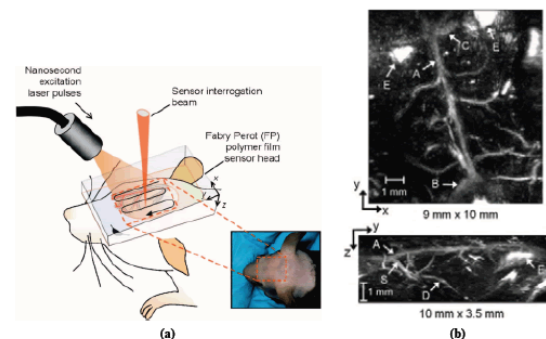
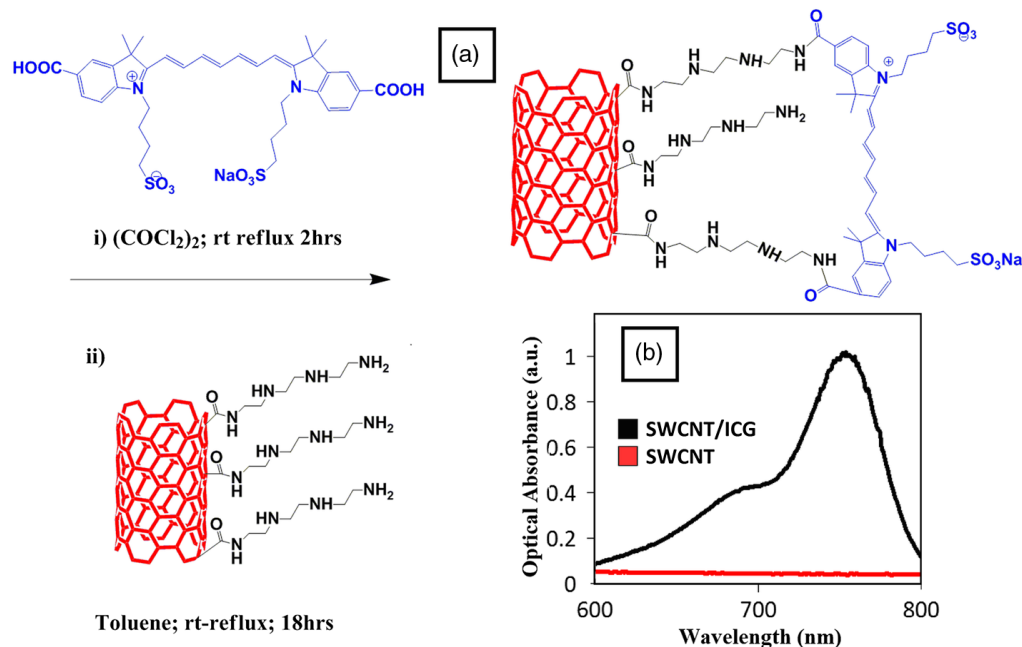
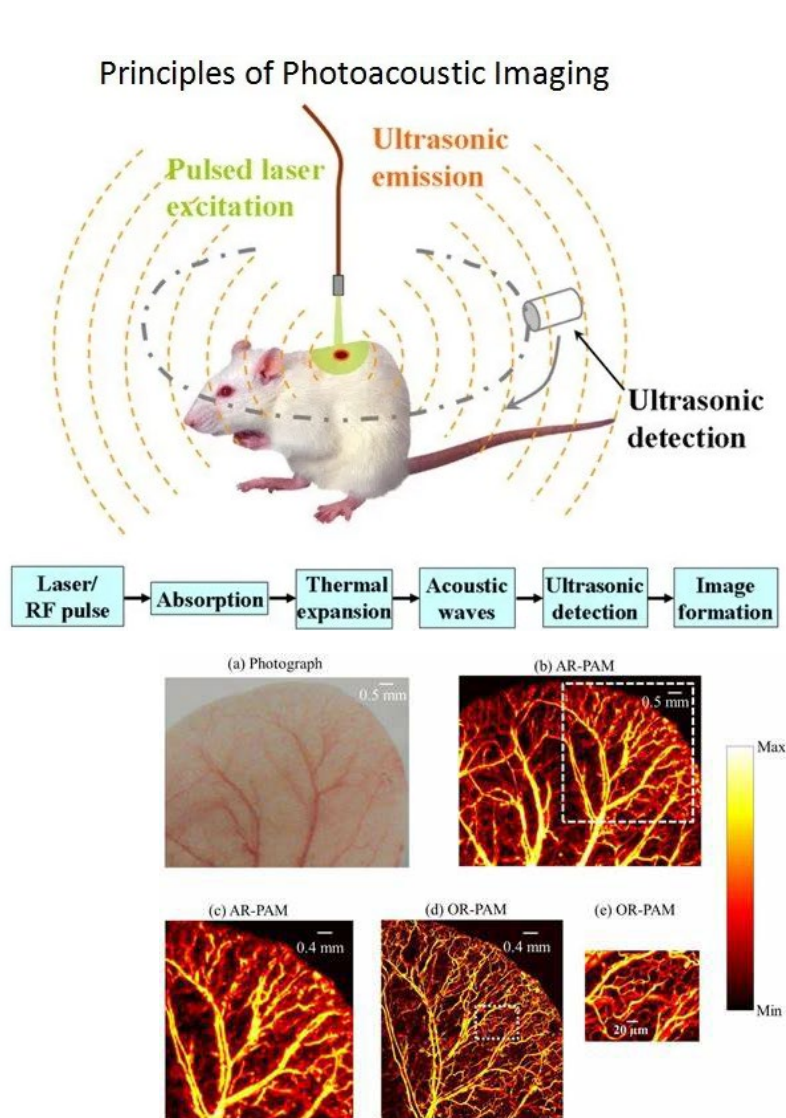


Figure 3: Experimental arrangement (a) and reconstructed images (b) in photoacoustic imaging of the vasculature in the mouse brain based on FP sensor. A, superior sagittal sinus B, transverse sinus; C—inferior cerebral vein; E, eyes; S, sinus rectus; D, inferior sagittal sinus. Reproduced with permission (Figure 1 and 4 of Reference 13).